The is the Optimal Way to Assess Symptomatic Fatigue in Degenerative Cervical Myelopathy: A Preliminary Study to Inform Content Validity

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Abstract

Background: Recent work has demonstrated that individuals with degenerative cervical myelopathy (DCM) often experience symptoms of fatigue which is under appreciated in the literature. Conversely, it is highly valuable to people living with DCM. The AO Spine RECODE-DCM study group recently determined no suitable fatigue scales exist for DCM.

Objective An assessment of content validity of existing fatigue scales used in other populations was undertaken to determine if any such scales are suitable for DCM.

Methods: An expert panel of people with DCM (PwDCM) was convened. PwDCM were members of the RECODE-DCM steering committee, ranged in severity from mild to severe myelopathy, and resided in the United Kingdom or the United States of America. 19 instruments in use for other pathologies were presented. A quantitative description of relevance to DCM was reported by PwDCM. Likewise, PwDCM scored each scale on clarity, and comprehensiveness. A preliminary qualitative analysis of comments on the various attributes of the respective instruments to establish themes distinguishing high-scoring instruments from lower-scoring instruments was assessed.

Results: The highest scoring scales were 1) modified fatigue impact scale for SCI (scored Relevance=3.78, Comprehensiveness=3.56, Clarity=4 respectively), 2) multidimensional assessment of fatigue (scored 3.67, 3.56, and 3.89 respectively), and 3) functional assessment of chronic illness therapy: fatigue scale (scored 3.56, 3.56, and 3.89 respectively). Common themes used in discussing various instruments were sleep, pain, psychosocial factors, and employment.

Conclusions:

The Modified Fatigue Impact Scale for Spinal Cord Injury, Multidimensional Assessment of Fatigue, and Functional Assessment of Chronic Illness Therapy-Fatigue Scale appear to be the most appropriate and validated instruments for assessing fatigue in patients with DCM. These scales demonstrate strong psychometric properties while addressing the unique symptomatic profile and functional limitations characteristic of DCM, enabling more accurate measurement and clinical monitoring of fatigue in this patient population.

Declarations of Conflict of Interest

Declarations of interest: none

Declaration of Use of Generative AI

No generative AI was used in writing or revising this manuscript.

Abbreviations

DCM – Degenerative cervical myelopathy
PwDCM – People with Degenerative cervical myelopathy
RECODE-DCM - REsearch objectives and COmmon Data Elements for Degenerative Cervical Myelopathy
mJOA – Modified Japanese Orthopedic Association
VAS-F - Visual analogue scale for fatigue
FAI - Fatigue assessment instrument
MFIS-SCI - Modified fatigue impact scale for spinal cord injury
SF-36-Fatigue - 36-item short-form health survey: fatigue subscale
FACIT-F Functional assessment of chronic illness therapy: fatigue scale
MAF - Multidimensional assessment of fatigue
CFS-APQ - Chronic fatigue syndrome-activities and participation questionnaire
FSI - Fatigue symptom inventory
CIS - Checklist individual strength
FFS - Fibrofatigue scale
BRAF-MDQ - Bristol rheumatoid arthritis fatigue: multidimensional questionnaire
MDF-fibro-17 - Multidimensional daily diary of fatigue-fibromyalgia-17 items
FAS - Fatigue assessment scale
NRS – Numeric rating scale: fatigue
CFS - Chalder fatigue scale
BFI - Brief fatigue inventory
FIS - Fatigue impact scale
MFI - Multidimensional fatigue inventory
PFS - Piper fatigue scale

Introduction

Degenerative cervical myelopathy (DCM) [1–4] is a common spinal cord condition whereby spinal intervertebral disc degeneration or hypertrophy and calcification of ligaments results in chronic, progressive compression of the spinal cord.[5] While many pathophysiological questions remain, the known similarity to spinal cord injury suggests an overlap of clinical features, including fatigue.[6] Fatigue has been identified as a core outcome for clinical trials on treatments for DCM by RECODE-DCM.[7,8] This process went on to recommend a list of outcome tools to measure the core outcomes. [10,11] As part of this process, we explored whether any existing fatigue scale should be included. After establishing the absence of DCM-specific scales, we reviewed candidate scales from other conditions to analyze their content validity in DCM.[9–11]

Potential physiological underpinnings of elevated fatigue in people with DCM (PwDCM) include weakness, reduction of corticospinal drive to lower motor neurons, pain, medication-related somnolence, inflammation, spasticity, sarcopenia, and reduction in physical activity.[12–16] In addition, reduced functional capabilities, potential risk of injury with injury with activities, and pain influencing sleep, can reduce physical reserve and worsen fatigue. To date, self-reported measurement scales for fatigue have not been used in PwDCM. Thus the content validity of potential fatigue scales is unknown in this patient population. Identification of patient-reported outcome scales for fatigue will buttress future investigations into objective assessments of neuromuscular fatiguability and may suggest roles in which physical therapy or other novel interventions may act to improve health-related quality of life.

Assessments of content validity are critical aspects for validating any self-reported outcome measure as per the COnsensus-based Standards for the Selection of Health Measurement Instruments (COSMIN) framework. [17–19] Specifically, if items are not relevant, comprehensive, and comprehensible to the target population and for the characteristic being studied, all other measurement properties (structural validity, internal consistency, repeatability) will be affected.[18] This indicates content validity is foremost in determining a tools appropriateness for use. COSMIN recommends that evaluation includes persons with lived experience of the disease.

Most clinician and patient-reported outcome measures proposed for DCM have not been adequately validated.[9] Critically, this lack of validation of outcome measures is likely to contribute to and perpetuate known research inefficiencies in DCM[20], such as the misalignment of research to patient priorities, increasing need to re-do trials due to variable outcomes, and inability to pool findings in future systematic reviews.

Therefore, this study's purpose was to provide a provisional assessment of content validity for existing fatigue using a panel of PwDCM. Tools were shortlisted from a systematic review[9]. These findings were principally used to inform the development of a DCM Minimum Dataset but can also serve as formative step for researchers investigating fatigue in DCM.

Methods

The 9 PwDCM who were asked to rate the survey scales were all members of the RECODE-DCM steering committee. RECODE-DCM is an international community working to accelerate knowledge discovery and translation that can improve outcomes in DCM. This original project was delivered in partnership with AO Spine, a global non-profit organization for spine surgeons. RECODE-DCM now sits within Myelopathy.org, a global DCM charity. Expertise in PwDCM was defined as having a diagnosis of DCM while also serving on DCM consensus workgroups, managing support groups, engaging in research, being involved in the Myelopathy.org charity, and / or other DCM patient advocacy efforts. Notably, the first author of this paper was a survey rater. His summary scores largely agreed with the prevailing views of the other raters. Scores including his data were highly correlated with scores excluding his data (r=0.98). Of the 9 PwDCM who rated scales, 6 provided demographic information. Of these 6 individuals, 4 were female, 3 resided in the United Kingdom whereas 3 resided in the United States of America, mJOA was 11.00±4.05 (range 7-17 out of 18 points), age was 53.38±12.14 (range 33.17 – 69.98 years).

The methods specific to this study have been previously reported elsewhere.[10] In brief, a systematic review was undertaken to identify questionnaire instruments related to fatigue used in research related to DCM. If no suitable scale was identified, the systematic review was expanded to conceptually similar conditions to DCM that affect the nervous system (i.e., traumatic spinal cord injury, fibromyalgia, stroke). Ethical approval was obtained from the University of Cambridge (HBREC2019.14) in accordance with the Declaration of Helsinki.

Purposive sampling was used to recruit PwDCM who were members of the steering committee for the RECODE-DCM project who were queried regarding their perceptions of 1) relevance, 2) comprehensiveness, and 3) clarity of the fatigue scales

(Appendix A). Question items and responses were adapted from the COSMIN manual. Perceptions were scored on four-point Likert scales, ranging from 1 (low intensity of agreement) to 4 (high intensity of agreement). For the Relevance item, 1 implied “no items are relevant” and 4 indicated “all items are relevant”. Responses to the Comprehensiveness question ranged from 1 “key aspects are missing” and 4 “no key aspects are missing. For, Clarity, 1 indicated “not clear” and 4 “very clear”. Operationally, we defined a mean score of ≥3.5 across raters as a “good” score for a respective domain. Additionally, PwDCM were asked to estimate how long (duration in minutes) each instrument would require for completion. This was considered of relevance to the projects principal purpose of informing a minimum data set for DCM research, where given a wide range of core outcomes, the potential burden on participants to complete all assessments would need to be balanced against its broader value for DCM research. Finally, a free response question was offered to PwDCM to solicit comments on the perceived qualities of each instrument. This was intended to provide preliminary evidence of the perspectives of PwDCM with respect to these potential fatigue scales listed below. This evidence will be used to streamline future mixed-methods studies. This cohort of individuals was derived largely from the United States and the United Kingdom. Following the scoring of each domain (e.g relevance, comprehensiveness and clarity), scores across domains were averaged to obtain a summary score from each person for each instrument.

In total, 9 PwDCM were asked to evaluate and provide scores for 19 fatigue-related tools. These tools were the: Visual analogue scale for fatigue (VAS-F)[21], Fatigue assessment instrument (FAI)[22], Modified fatigue impact scale for spinal cord injury

(MFIS-SCI)[23], 36-item short-form health survey: fatigue subscale (SF-36-Fatigue)[24], Functional assessment of chronic illness therapy: fatigue scale (FACIT-F)[25], Multidimensional assessment of fatigue (MAF)[26], Chronic fatigue syndrome-activities and participation questionnaire (CFS-APQ)[27], Fatigue symptom inventory (FSI)[28], Checklist individual strength (CIS)[29], Fibrofatigue scale (FFS)[30], Bristol rheumatoid arthritis fatigue: multidimensional questionnaire (BRAF-MDQ)[31], Multidimensional daily diary of fatigue-fibromyalgia-17 items (MDF-fibro-17)[32], Fatigue assessment scale (FAS)[33], Numeric rating scale: fatigue[34], Chalder fatigue scale (CFS)[34], Brief fatigue inventory (BFI)[35], Fatigue impact scale (FIS)[36], Multidimensional fatigue inventory (MFI)[37], Piper fatigue scale (PFS)[38].

In addition to providing subjective scoring, respondents were asked to provide free text responses, via an online survey, to discuss what they did/did not like about respective tools. Responses and results were returned to participants for comment. The survey was sent to participants via email from one investigator (BMD) a male trainee physician trained in qualitative methods. Data was collected virtually. These responses were analyzed qualitatively to identify common themes. Free responses were qualitatively analyzed to condense similar comments into various themes. As such, questions broadly related to the length, layout, or clarity of the instrument were re-coded to “structure.” Comments including words such as psychosocial, socioemotional, etc. were re-coded “psychosocial”. Comments related to the length of recall were re-coded to “recall”. Filler words such as “liked”, “mentions”, “perhaps”, etc. were discarded. Themes were identified from the data and responses were stored in Excel. All of the free responses were then entered into word cloud generating software which extracted the top 10 most frequent words, which is a validated method.[39,40] This was intentended to inform future qualitative focus-groups studies.

Statistical analysis

Given the qualitative-to-semi-quantitative nature of the present study, no formal hypothesis testing was undertaken. Nevertheless, descriptive and summary statistics have been calculated in Excel (Microsoft, INC., Redmond, WA, United States).

Results

When pooled across all scores from all participants for all instruments, the median summary score was 3.67, indicating raters scored all instruments rather well. Four instruments had a between-participant average >3.67 (MFIS-SCI, FACIT-F, MAF, and FAI). Two of the scales received a summary score <3, specifically FFS (summary score = 2.54) and MDF-fibro-17 (summary score = 2.96). Generally, across scales, Comprehensiveness was scored lower than Relevance and Clarity. Scores for the various scales are presented in Figure 1. The frequency of themes with respect to various scales are visually presented in Figure 2.

Time to Complete

Of the top-scored scales, the, mean times to complete were around 6 minutes (MFIS-SCI=6.00±4.03 mins; FACIT-F=7.56± 3.97 mins; MAF= 6.67±3.46 mins). Thus, most scales could easily be administered without adding substantial testing burden in many clinical trials.

Preliminary Qualitative Analysis of Content Validity

Several prevailing themes emerged from the comments received from the focus group. One prevailing theme was an insufficient discussion of sleep. This was noted in regard to the FIS, the MFIS-SCI, MAF, BFI, and MFI. About the FIS some specific comments included, "*the problem with the fatigue I experience is that I have to sleep during the day 3 - 4 hours & the sleep I have at night is never refreshing & often disturbed [participant 7]."* Another responder suggested questions be added to the FIS and MFIS-SCI about requiring naps, such as, "*Do you require a nap during the day [participant 9]?*". Considering the MFIS-SCI one responder said, "The impact of sleep or sleep disturbance is missing [participant 7]". Another comment noted the BFI "Doesn't include pain or sleep [participant 7]". On the other hand, the FACIT-F was commended for discussing sleep such as, "I like this one even better because it mentions the socioemotional impact of illness on yourself & the family & mentions sleep. It mentions pain too [participant 7]."

Other common themes were related to the effects of pain on fatigue and fatigue on employment. For example, both the MAF, MDF-Fibro-17, MFI, CIS and the FFS were criticized for not accounting for the effects of pain on fatigue and/or fatigue on employment status. FACIT-F and MFIS-SCI, however, were commended for mentioning pain. For the MFIS-SCI it was said, "It discusses physical discomfort which is good [participant 7]".

Likewise, mood/psychological state was also a topic frequently commented on. FACIT-F and SF-36 Quality of Life were commended for discussing "socioemotional impact". Other scales, such as the FAS and CIS, were criticized on this topic.

Other frequent comments related to the topic of aesthetics and setup/structure of the survey. This includes comments such as "Far too detailed [participant 3]", "Feels like duplicate questions [participant 5]", "*It would take me [a long] time to recollect all the information it asks for[participant 7]*" for the FSI and "*A bit long winded for a person with DCM to fill out*" for the PFI. This topic also included comments that the responder was "*distracted/bothered by the layout or dark highlighted bars [participant 4]*" for the BFI.

Another common topic of comment was on the duration of recall required from respective scales. For example, scales that asked about symptoms within a 24-hour period or "right now" were viewed unfavorably, compared to those that asked about symptoms within a 2-4 ereweek period. For example, participant 8 commented regarding the FBI "24 hours seem like such a short time frame". Another participant [participant 9] wrote regarding the BFI, "I am not sure 24-hour snapshot is sufficient. We have good and bad days". This sentiment was related to views that DCM is variable and that too brief a recall time period would fail to capture the worst days. For example, regarding the CFS, one person wrote "*This is a months snapshot, this one works well.* [participants 9]"

Common Criticisms of Poor versus High Performing Scales

In analysing the criticisms of the bottom three performing scales, length/detail, format, and extraneous concepts were frequently noted for the FSI, CIS, and FFS. For example, criticisms of the FFS frequently included its long, poorly designed layout. Comments on the CIS included “Not as easy as the other formats [participant 2]” and “confusing – the scoring [participant 4]”. Likewise, comments on the FSI includedincluded ”Misses out on the physical fatiguability possible in DCM [participant 1]”, “far too detailed [participant 3]”, “would dwell on these [participant 4]”, “It would take me a long time to recollect all the information it asks for.[participant 7].” Conversely, despite sometimes being criticised for being long, the top-scoring scales criticized were described as more “clear” but also “relevant,” “comprehensive,” and “to the point.” Regarding the MFIS-SCI, comments included “Nicely broken down and helpful [participant 4]”, “clearer questions and less bothersome [participant 5]”, “I like this scale – it seems comprehensive [participant 7]”. Similarly of the FACIT-F, one person responded, “this one seems more well-rounded [participant 8]”. Regarding the MAF, one person responded, “I found this instrument a bit more to the point [participant 5]”.

Discussion

The results of this study suggest that several fatigue scales from the broader literature would be considered acceptable for research on fatigue in PwDCM. The highest performing scales among PwDCM were the MFIS-SCI, MAF, and FACIT-F, which could represent starting points to implement a fatigue scale into use in research in DCM. Our present results further suggest that any future development of fatigue scales specific to DCM should incorporate assessments of sleep, effects of pain on fatigue, psychosocial factors such as stress due to suffering from a chronic condition, and impact of fatigue on employment status. Alternatively, additional assessment scales for pain (e.g. neck disability index, brief pain inventory, numeric rating scale for pain), employment status and mental health could supplement deficiencies in assessments of fatigue on these factors. These are in addition to assessments of lower extremity fatiguability (such as 6-minute walk test) that have been shown to differ from healthy controls.[41] Further, future studies examining the impact of symptomatic fatigue and neuromuscular fatigability on quality of life are needed.

As noted above, fatigue was identified as a core outcome of DCM; however, we have little understanding of its biological basis in DCM. This presents several issues related to measurement validity. PwDCM have identified sleep, pain, employment, and mental health as common critical components of fatigue in DCM. While the overall relationship of many of these to fatigue is poorly understood (and this includes causal direction), we discuss them subsequently and present a hypothesized formative model of poor sleep, pain, and psychosocial factors driving fatigue, with employment status likely a consequence of fatigue.

The MFIS-SCI is the least surprising scale to be relevant to PwDCM as it is drawn from a highly similar clinical population (e.g. SCI vs. DCM) rather than chronic fatigue syndromes. These factors suggest that the MFIS-SCI is the most likely to effectively translate to a DCM population. The MFIS is also validated in multiple sclerosis and is designed to capture the “impact” of fatigue.[42,43]

Critical themes: Sleep

Sleep disturbances reportedly affect approximately 70% of PwDCM[44] and are also common in individuals who sustained a traumatic SCI. The consequences of disruption to the circadian system and sleep can be profound and include myriad metabolic ramifications with widespread health consequences .[45] Physiological changes in this theme circadian rhythm have been demonstrated in individuals with spinal cord lesions[46,47] but not yet studied in DCM. Thus, a heightened interest in this by the respondents is unsurprising. Sleep disturbances can be profoundly disruptive to health-related quality of life and limit social participation and employability.[48] A recent investigation of sleep deprivation in DCM identified worse depression scale score, a lower modified Japanese Orthopedic Association score, chronic shoulder joint pain, smaller spinal cord area, and decreased cervical range of motion as independent risk factors for sleep disturbance.[49] Critically, sleep disturbances are also intimately coupled with other themes discussed below, such as depression and pain.

To that end, the Neck Disability Index includes an item on the relationship between neck pathology and impaired sleep. Nevertheless, only a handful of studies have examined the relationship between DCM and poor sleep. As such, future research is needed to understand the potential impact of disordered sleeping on fatigue and functional impairments in PwDCM.

Critical Themes: Pain

The importance of the role of pain on fatigue is unsurprising. Previous works have determined that pain is a key recovery priority of PwDCM. In PwDCM who have pain, pain can be highly disabling, widespread, and either musculoskeletal or neuropathic in origin.[13] This mirrors other conditions such as fibromyalgia, where pain and fatigue are coupled.

Using the Neck Disability Index to obtain prevalence and severity scores of pain, pain may be present in approximately 80% of patients with DCM pre-operatively. Moreover, of those who had pain, approximately 80% were of moderate severity or worse[13,50] , and about 50% experienced pain that substantially impacted major life events.[51] Importantly, pain was more frequent in females and in patients ≤57 years old, with a body mass index ≥27, and with gastrointestinal and rheumatological diseases.[50] Moreover, DCM while known to cause frequent neck pain and occasional headaches, it can also have a widespread, impacting on all four limbs, the back, and the abdomen,.[13] Pain was commonly rated as interferential and weakly to strongly negatively associated with quality of life.[13] Beneficially, surgical interventions appear to reduce Neck Disability Index scores and transition a patients away from being classified as having high-impact chronic pain.[50,52]

Critical Themes: Psychosocial

Mental health is known to be substantially compromised in PwDCM,[53] and commonly associated with otherotherother factors such as pain and fatigue,[13] as highlighted in a recent focus group for PwDCM.[54] Indeed, the prevalence of depression in PwDCM was estimated to be 37%, close to that of lumbar disc herniation.[55] Fortunately, mental health related symptoms appear to improve following surgical decompression.[56] Having described the mental health experiences of PwDCM, the interrelationships between mental health, pain, and fatigue are discussed in the section below on the biopsychosocial model.

Critical Themes: Employment

In other conditions, fatigue is frequently negatively associated with employment outcomes. For example, a recent systematic review on cancer survivors identified 38 studies examining fatigue in the context of employment and productivity, and concluded that employment status and productivity were both negatively affected by fatigue.[57] Likewise, negative associations between increased fatigue and employment outcomes have classically been documented in fibromyalgia, multiple sclerosis, and, more recently, COVID-19. Conversely, recently, fatigue has been reported as more of a barrier to employment for people with multiple sclerosis than spinal cord injury.[58] It is presently unclear how strongly fatigue may impact employment status and productivity in PwDCM. As other scales may over represent the importance of fatigue, this would further support the adoption of spinal cord injury-specific fatigue scales (e.g.,.,., MFIS-SCI), except that the MFIS-SCI does not explicitly mention employment.[59] The MFIS-SCI does discuss more broadly the impact of fatigue on tasks, activities, social activities, and lifestyles in which respondents may include employment. This was not discussed by our panel of PwDCM with lived experiences. Thus, broader consensus on the applicability of these questions to employment is unknown. Further research is therefore necessary to identify if these items of the MFIS-SCI are associated with employment status, specifically in PwDCM.

Critical Themes: Structure

There were frequent criticisms of the length, layout, or clarity of the surveys that were evaluated; this speaks to the overall testing burden on the participant to complete such surveys. Surveys that are overly long or convoluted will be cognitively fatiguing. If future instruments are developed to assess fatigue tailored to DCM, more focused questionnaires are likely needed. Likewise, concise, easily understandable questions arranged in an aesthetically logical and appealing way will all be critical. In that vein, “clarity” may have a protective effect on testing burden. That is, the top-scoring scales, even though they were sometimes lengthy, scored well in part because our focus group described them as clear and “to the point”. This would suggest that participants may be willing to take part in longer scales (more questions) if those scales are well written such that there is a lower cognitive demand per question.

Biopsychosocial Model

Improving our understanding of the pathophysiology of DCM is critical. Likewise, furthering our understanding of the pathophysiological mechanisms leading from spinal cord compression to symptoms such as fatigue will guide novel therapies to alleviate symptom burden in patients. As noted above, several of the themes identified by our expert panel of PwDCM are interrelated and consistent with other fatigue-related conditions. Themes such as sleepiness, pain, and mental health have been richly documented in fibromyalgia. It is not known, however, the extent to which the causal directionalities of these constructs are similar in DCM as they are in other conditions.

Together, this cluster of symptoms is described using the biopsychosocial model (Figure 3), which has been applied to appreciate the interrelationship between pain, mental health, and fatigue in a variety of other conditions, including chronic fatigue syndrome, fibromyalgia, cancer, traumatic brain injury, and others. Importantly, there is preliminary evidence in PwDCM for a number of factors within the biopsychosocial model that contribute to fatigue (such as pain, inflammation, endurance, and mental health, [among other studies]).[6,13,16,41,53]

In cancer, for example,[60] fatigue was hypothesized to be caused by reductions in strength, endurance, cardiopulmonary fitness, and body composition as physiological factors. Likewise, fatigue was proposed to be directly caused by changes to immunological, metabolic, endocrine functions, and systemic inflammation. Fatigue was proposed to be caused via indirect pathways made up of psychological (e.g. depression, anxiety, distress and congition), behavioral (e.g. sleep quantiy and quality, and appetite), and social (e.g. social interaction and positive reinforcement) factors. ,While this description above from McNeely and Courneya[60] did not include pain, other relevant descriptions of biopsychosocial models of fatigue, such as for rheumatoid arthritis,[61] add pain to their model.

Limitations

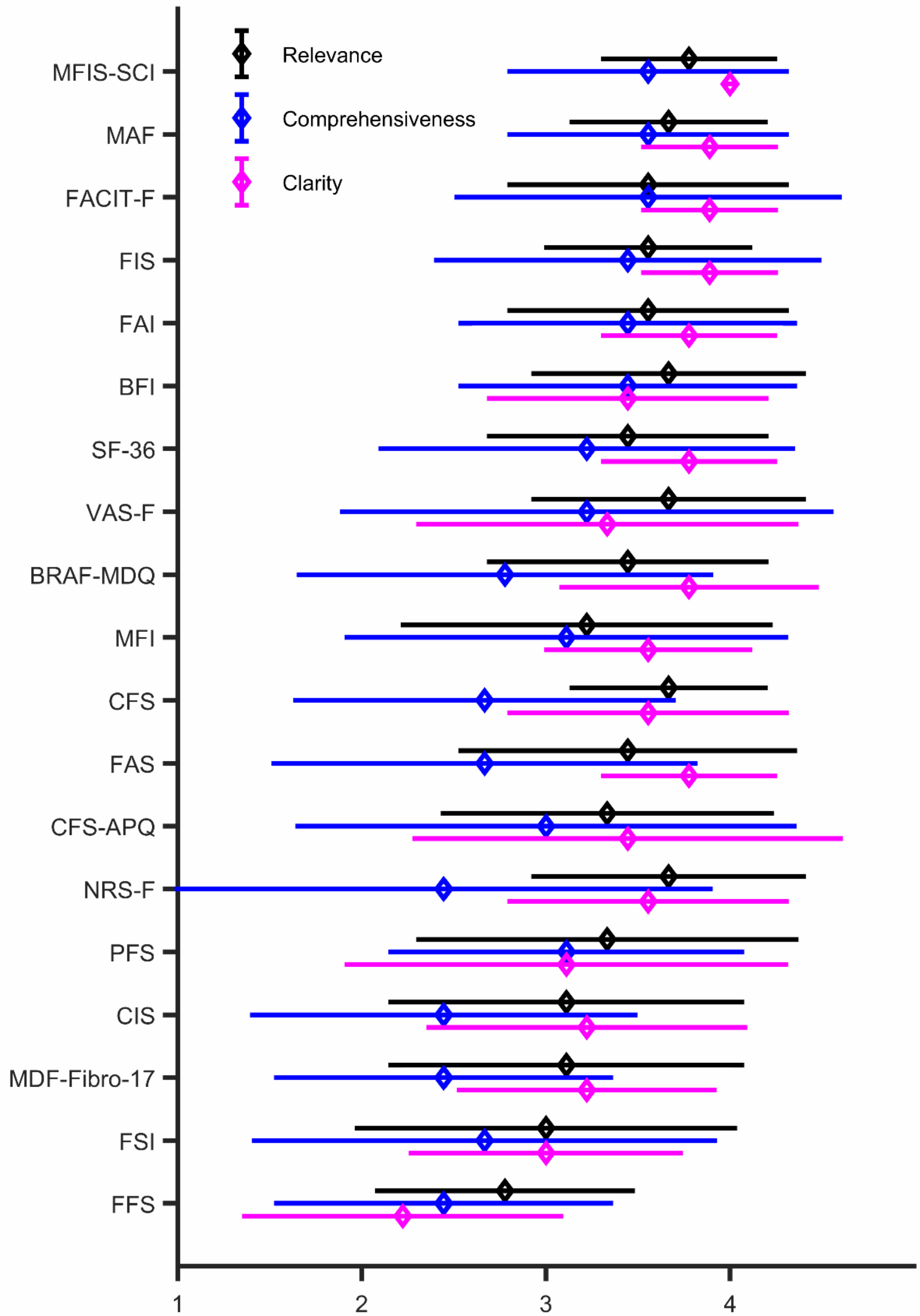
This study examined the perceptions of people living with DCM regarding the appropriateness of various existing self-reported outcome scales assessing fatigue. One potential limitation of this study was the small sample size ,potentially resulting in a failure to reach saturation. However, our sample of PwDCM was recruited from multiple countries across Europe and North America and consisted of seasoned advocates, who served on the RECODE-DCM steering committee and led peer support groups educating and advocating for PwDCM. Given these backgrounds, our panelists were well suited to gauge the included fatigue surveys based on their own experiences as well as common questions and symptom complaints raised by others. A follow-up, true qualitative study using focus groups, should be conducted based on the short-list scales to recruit additional opinions. Another potential limitation is related to cross-cultural adaptability.[62] All scales were tested in English and participants surveyed here were all native English speakers. While English makes nuanced distinctions between words such as “sleepy”, “tired”, and “fatigued”, other languages make no distinction between some of these terms. For this reason, our recommendations should only be taken for English-based studies unless there is a validated version in a research study’s target language (for example MFIS). Finally, numerous components exist for establishing the validity of an instrument. Other aspects of validity (structural, convergent, internal consistency, repeatability, et cetera) remain to be established in DCM specifically for these scales.

Conclusions

Fatigue is recognised as an important symptom of DCM, but it remains poorly understood. Existing measures are considered reasonable for use in DCM but need well-constructed field testing. Patients report that fatigue interacts with other experiences, which may be relevant when measuring and understanding this variable.

The MFIS-SCI, MAF, and FACIT-F were the preferred outcome measures according to the PwDCM who participated in this study.

Future studies will be required to assess the impact of fatigue on individuals with DCM and to evaluate therapeutic interventions to address this important symptom.



Relevance
Comprehensiveness
Clarity
MFIS-SCI
MAF
FACIT-F
FIS
FAI
BFI
SF-36
VAS-F
BRAF-MDQ
MFI
CFS
FAS
CFS-APQ
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Figure 1. Content validity scores for

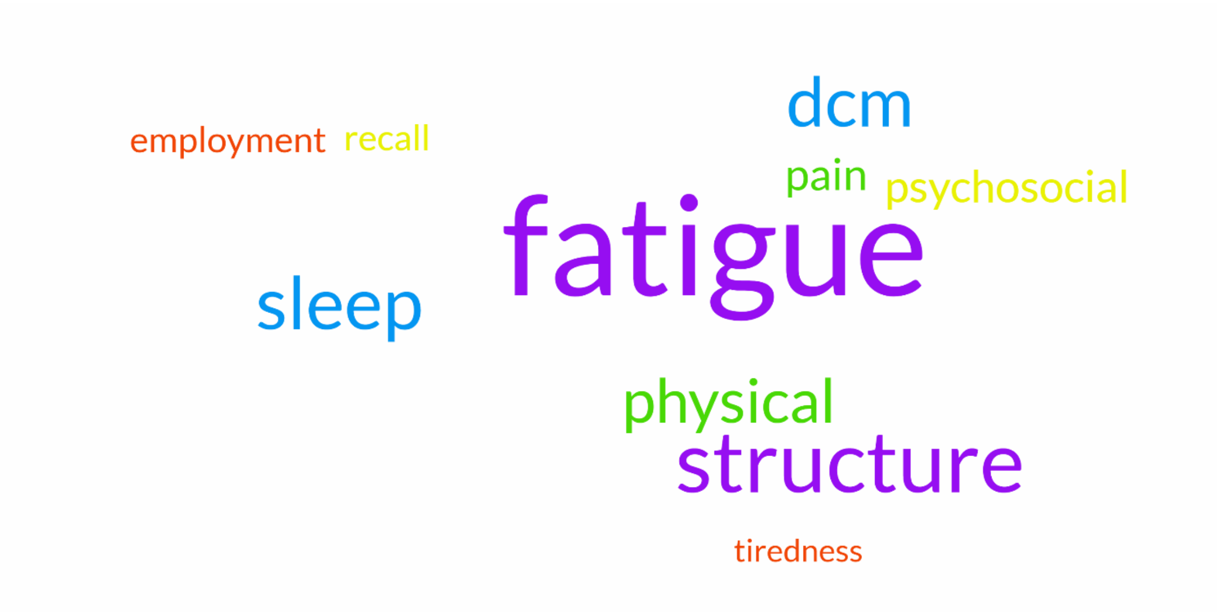

Figure 3. Word cloud based on responses. Words were limited to the top 10 words with respect to frequency.

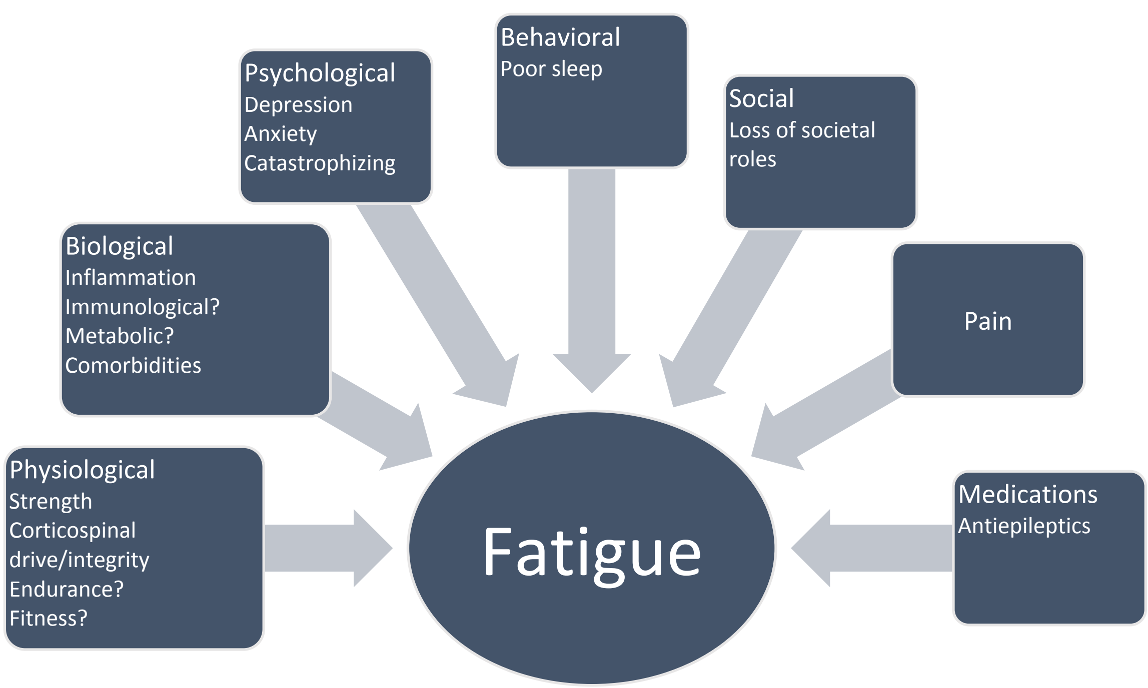

Figure 2. Biopsychosocial model applied to fatigue. This figure highlights the various biological, psychological, and social factors related to fatigue relevant to degenerative cervical myelopathy.

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